Design and Development of a Self-Righting Airdrop Floating Pod for Flood Relief

Introduction

As global warming and rapid urbanization continues, abnormal weather phenomena such as localized downpours and flooding have become more and more frequent and intense [1,2]. A great number of metropolitan cities have suffered from flood hazards, which incurred irreducible and severe damage to local people and urban infrastructures. During floods there are few major issues and challenges such as flood preparations, lack of rescuers, lack of awareness, communication difficulty, lack of assets for rescue, transportation issues, the absence of food supply, and other constraints. On the other hand, the challenges are lack of provisions, redevelopment, the social and economic situation, and trauma. The focus of our study is to tackle the key pressing issue of transportation. A major issue for transport is that there is no helicopter landing pad to send food in most flood affected areas. The only suitable option remaining is to throw the food from the air. The poor working condition of some of the available boats does not allow them to continue their rescue efforts. I think these needs to be emphasized and there should be a specific boat for the rescue operation. This is very important because the boat and hovercraft are the needed transport when the road is closed.To address these transportation constraints, our project proposes the development of a **remote controlled Self-righting Airdrop floating pod prototype** designed specifically for emergency supply delivery in flooded environments. In contrast to conventional boats or helicopters, this small, remotely operated vehicle is capable of traversing narrow waterways filled with debris without demanding a landing zone, allowing for extreme flexibility in urban flood situations. Together with these features, the water drone aims to provide a reliable, low-cost, and rapidly accessible resource. Successful deployment of this water drone could lead to significant reductions in time to deliver aid, improve the work of disaster-response teams, and ultimately save lives in serious flood events.

Literature Review

Self-righting vessels have been a focus of research for maritime and disaster-relief engineers for many years. Numerous research efforts have focused on the fundamental phenomena of buoyancy, shape of hull, and the mechanisms of buoyant stability. In reviewing four studies, we find pertinent references to the design and development of a self-righting flood relief pod with paddle-wheel propulsion.

Yin, Tomita, and Shavelson[1] made progress on addressing student misconceptions about floating and sinking events. While this work is geared toward teaching science students, it supports fundamental concepts in hydrostatics, particularly regarding center of gravity and center of buoyancy as important contributors to stability. They indicated that small changes in density distribution, effects from free surfaces, and internal movement of water would have a direct influence on an object remaining upright or capsizing. For the design of a relief pod, these concepts lead to considerations on ballast addition and placement, which sealed buoyancy chambers were good design ideas, and uncontrolled sloshing of water could produce instability.

Akyıldız and Şimşek [2] studied the concept of designing a self-righting boat. They highlighted two dominant strategies: designing the hull and deck profiles geometrically to create restoring moments, and designing low ballast to lower the CG. Their research indicated that a rounded hull or tumblehome hull form combined with weighted locations on the keel would provide passive self-righting capacity. As with previous studies cited, trade-offs such as depth and payload could not easily be discounted. This study supports the idea that a flood pod should rely on passive self-righting elements instead of complex active self-righting systems, due to the unpredictable nature of disaster conditions.

Drobyshevski [3] summarized studies of boats that upright from a floating position with their hull upside-down; it relates the buoyancy distribution on a hull in an inverted position with trapped air volume and the repair torques associated with the distribution of buoyancy. This research shows that hull form and compartmentalization are important aspects which may or may not support the ability of a boat to self-right; in a small boat even minor curvature differences will influence the self-righting ability. Furthermore, a trapped air volume may help support self-righting and a free surface may or may not support self-righting. It should be noted that in all cases examined, the initial condition of the hull influences whether a boat can self-right. These findings further support the idea of using sealed buoyancy compartments in the relief pod to help with self-righting when capsized.

Trimulyono et al. [4] discussed experimental and computational techniques for testing small-scale marine structures for their most recent study. They utilized model testing, computational fluid dynamics (CFD), and environmental modeling to determine righting moments, maneuverability, and resistance related to environmental loading. They concluded that when moving a theoretical design into practice, it was important to validate simulation and physical testing. Consequently, the methods discussed in the study by Trimulyono et al. will be particularly useful for evaluating recovery from capsizing, propulsion in shallow water, and survivability in a debris-laden environment for the flood relief pod.

These studies collectively provide direction for design requirements. Accurate hydrostatic analysis will be important for ensuring positive righting moments throughout the range of heel angles. Second, passive self-righting methods, relying solely on hull shape and the location of low ballast, provide the most robust and maintenance-free alternative for utilizing in an emergency. Third, there should be a sealed buoyancy compartment or compartments included in the design that contribute to stability and potentially recovery. Finally, validating the design with trial experiments - scaled in the laboratory and/or full-scale with the flood conditions - will have to take place to demonstrate effectiveness. This information strongly supported the prototype flood relief pod design which combined self-righting characteristics with paddle-wheel propulsion to create a resilient and maneuverable craft in response to emergencies.

Methodology

The proposed prototype was designed as a lightweight flood-relief pod with an intended mass of **2–3 kg** and a payload capacity of approximately **0.5 kg**. The overall dimensions of the pod are **355 × 130 × 90 mm**, with the hull thickness set at **5 mm**. To achieve passive self-righting capability, a **solid steel ballast bar** of dimensions **25 × 25 × 100 mm** was incorporated at the base. The primary structure of the hull was fabricated from **PLA material** using **3D printing with 30% infill density**, ensuring a balance between structural integrity and reduced weight.

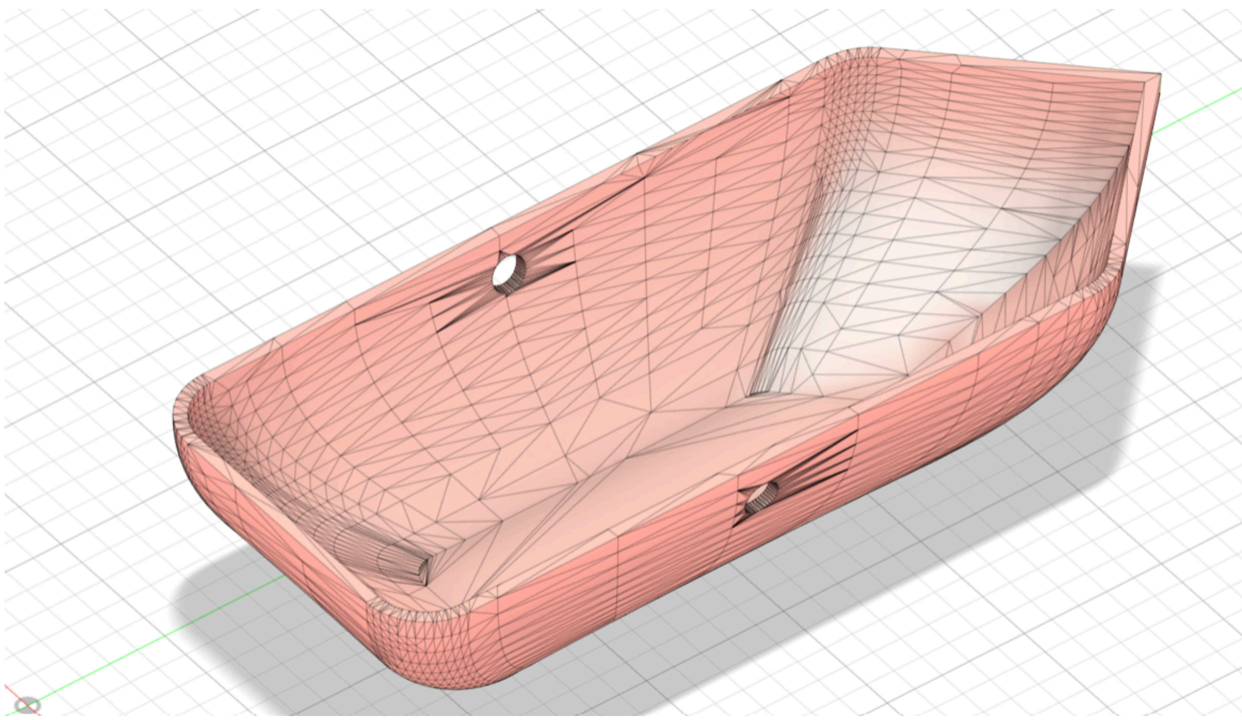

Fig 1: Boat iteration CAD

The physical prototype of the relief boat has been built for stability, waterproofing, and functional maneuverability. This boat is equipped with paddle wheels, which are composed of two 3D-printed assemblies, one mounted on each side of the hull and driven by dual motors instead of traditional propulsion. These paddle wheels are operated using SG996 high-torque servo motors, which have been securely mounted inside the hull to prevent any exposure to water damage. The hull has also been sealed with silicone paste to waterproof it as well as the electronics used to operate the boat in flood zone conditions.

The electronic control box is mounted at the top of the boat, containing electronics for the ESP32 microcontroller, motor driver electronics, and power supply. The front of the boat contains a round lid that can be unscrewed to allow easy access for maintenance, but also to allow for operational flexibility such as loading small relief items onboard or even servicing the servo if

needed. There is also a steel ballast that has been secured below the hull of the boat that has provided a lower center of gravity to improve stability against overturning in choppy waters.
Overall, this design provides functionally robust propulsion and waterproof protection along with storage capacity for small relief supplies making the relief boat prototype suited for navigation across flooded areas.

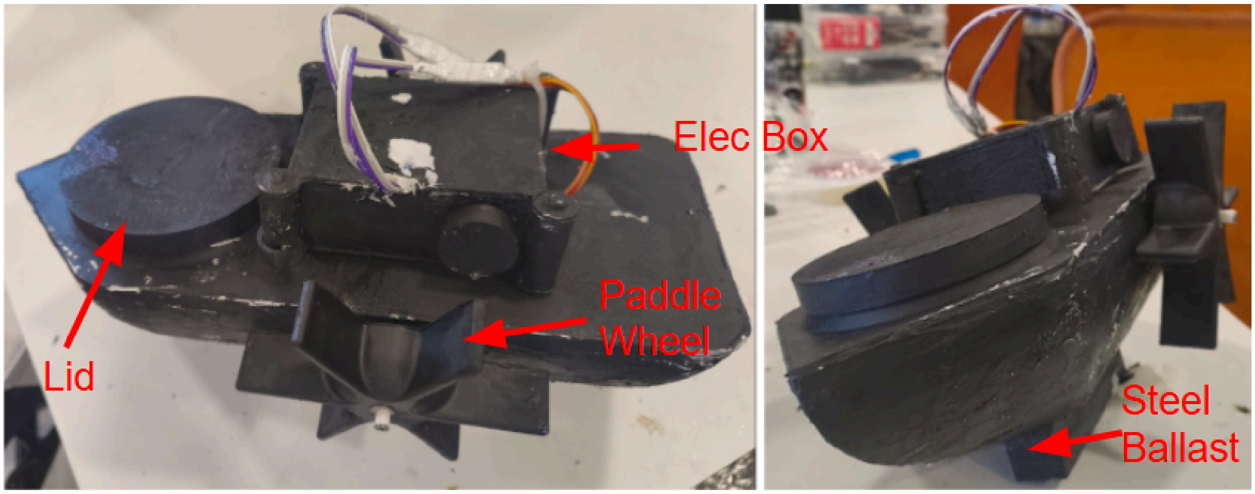

Fig 2: Physical prototype of the relief pod

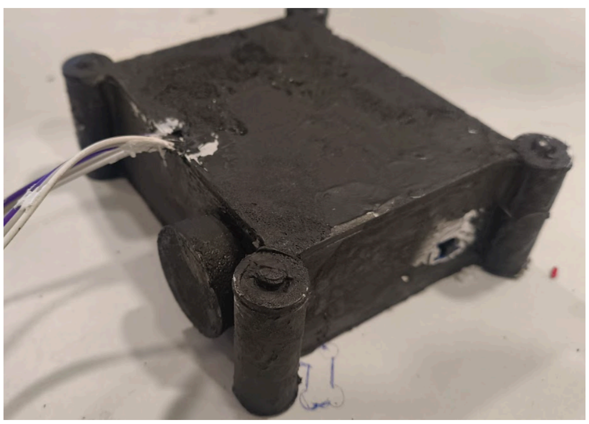

Fig 3: Electronic box

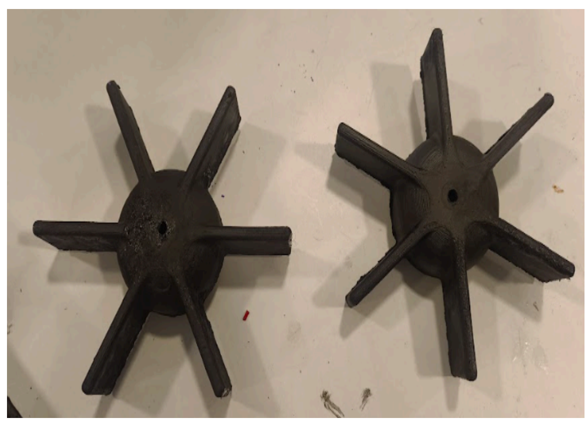

Fig 4: Paddle Wheels

Buoyancy Considerations

According to **Archimedes' Principle**, the buoyant force acting on a floating body is equal to the weight of the displaced fluid [2]:

$F_b = \rho_w \times g \times V_d$

Where:

- F_b = Buoyant force (N)
- ρ_w = Density of water ($\approx$1000 kg/m^3)
- g = Acceleration due to gravity (9.81 m/s^2)

- V_d= Volume of displaced water (m^3)

For stable floating, the condition is [2]:

$W_{total} \leq F_b$

Where W_{total} is the combined weight of the boat and payload.

Volume Displacement

For calculation, the pod is considered to submerge to a draft of **50 mm** under no external payload:

$V_d = L \times B \times T$
$V_d = 355\ mm \times 130\ mm \times 50\ mm,$
$V_d = 2{,}307{,}500\ mm^3 = 2.31\times10^{-3}\ m^3$

Thus, the buoyant force is:
$F_b = \rho_w \text{ x } g \text{ x } V_d = (1000)\ (9.81)\ (2.31\times10^{-3}) = 22.65\ N$

Corresponding mass of displaced water:

$M_{disp} = F_b\ /\ g = 2.31\ kg$

This represents the maximum weight that can be supported at 50 mm submergence.

Payload Capacity

Assuming the hull weight is approximately **1.8–2.0 kg** (PLA structure + ballast + electronics), the available payload capacity is:

$M_{payload} = m_{disp} - m_{hull}$
$M_{payload} \approx 2.31\ kg - 1.8\ kg = 0.5\ kg$

This matches the design requirement of carrying **~500 g of supplies**.

Stability Consideration

The placement of the **steel ballast bar** at the bottom lowers the **center of gravity (CG)**, ensuring a righting moment when the pod is tilted. The restoring moment Mr can be expressed as [2]:

$Mr = \Delta\ GM \sin\theta$

Where:

- Δ = Displacement weight of the pod (N)
- GM = Metacentric height (distance between center of gravity and metacenter)
- θ= Heel angle

By ensuring $GM > 0$, the prototype achieves **positive stability**, enabling self-righting within a heel range of −90° to +90°.

Boat Electronics

The onboard electronics of the relief pod (Figure 5) centered around a ESP32 microcontroller responsible for wireless communication, control, and actuator. The primary power source is a 3.7 V Li-Po battery, which recharges using a TP4056 charging/protection circuit board. An on/off switch isolates the system, while the DC–DC boost converter raises the voltage of the battery to 5 V, providing regulated power to both the ESP32 and servos. All grounds (battery, converter, ESP32 and servos) connect to a common reference for stable operation.

Two servo motors connect directly to the ESP32 using PWM pins. The servo motors actuate the paddle-wheel mechanism, allowing for forward, reverse, and turning motion. Because hobby servos rely on a 3.3 V logic signal, no level shifting is necessary between the GPIO and servo input. To stabilize power during rapid torque demands on the servos, a 470 µF electrolytic capacitor is placed close to the servos. This is further supported with 0.1 µF ceramic capacitors placed near the ESP32.

The ESP32 receives control commands from the handheld remote using either the Wi-Fi or the Bluetooth capability it has, translates that command to appropriate PWM outputs. The average draw is approximately 0.8 A at 5 V (with 0.2 A from the ESP32, and 0.3 A for each of the two servos). Using an 150 mAh Li-Po battery has limited battery life to approximately 7-10 minutes. This indicates the necessity for larger batteries for scaled prototypes. Lastly, with the compact electronic system, the modular design, and lightweight design, it was able to navigate in flood conditions while maintaining the pod's self-righting design.

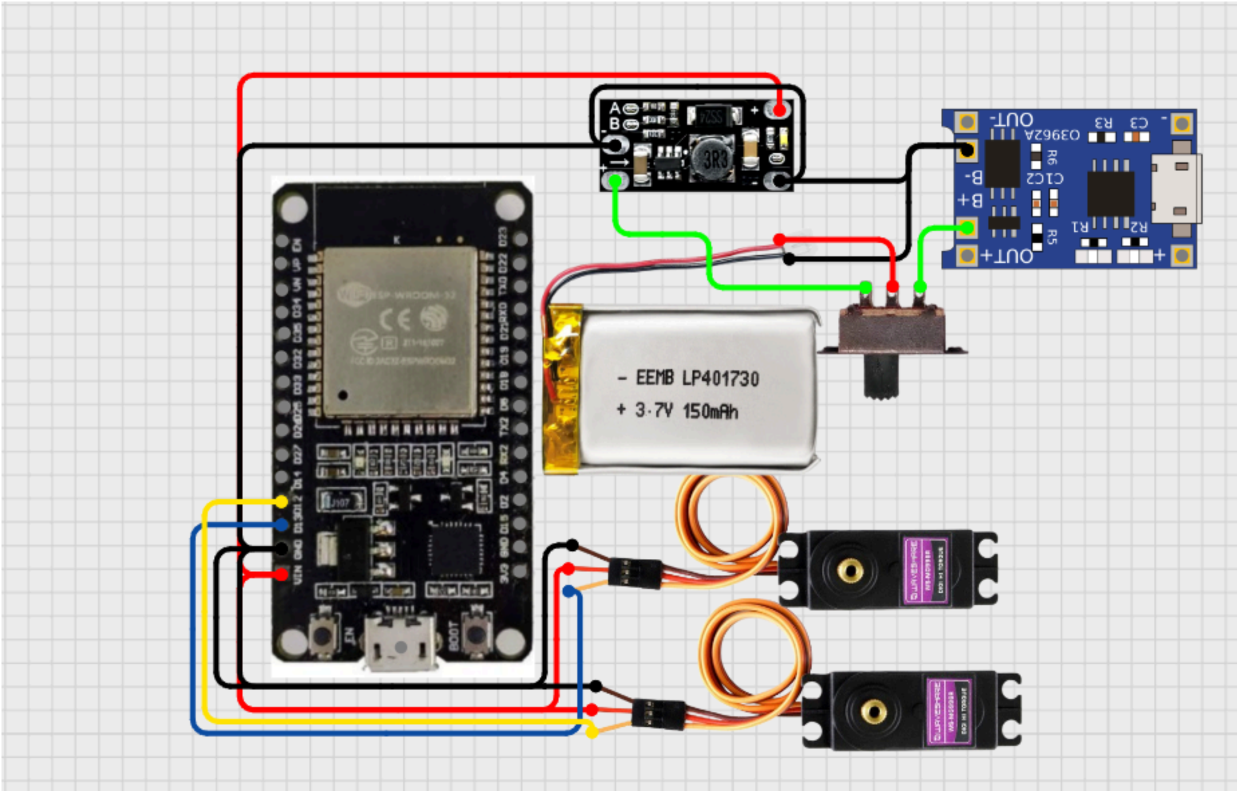

Fig 5: Boat onboard electronics circuit diagram

The remote control device illustrated in Figure 6 is assembled around an ESP32 microcontroller , powered by a 3.7 V Li-Po battery managed and protected by a TP4056 module. An on/off switch disconnects the battery. A DC-DC boost converter increases voltage to a stable 5 V, which is input to the ESP32 Vin pin. Four momentary push buttons are used as directional inputs (forward, backward, left, right) and each push button connects to a dedicated ESP32 GPIO pin with a 10 kΩ pull-down resistor to maintain stable logic levels. When pressed, the push button drives the pin HIGH and the ESP32 firmware interprets this as a navigation command for the unit.

The ESP32 watches the button inputs at all times, encodes the commands, and sends them to the onboard ESP32 wirelessly via Wi-Fi (ESP-NOW) or Bluetooth. The system uses software debouncing routines to eliminate false triggers, enabling smooth and reliable control. The system draws approximately 200-220 mA during active transmission, resulting in total power consumption of approximately 1 W. A 150 mAh battery, with regulator efficiencies of ~90%, implies operational times of 30-40 minutes of continuous use, ideal for testing, although a larger battery would be recommended for field deployments.

For the build, the Li-Po cell, TP4056, and boost module are installed inside a handheld unit with buttons positioned ergonomically on the case. The 5 V supply is decoupled with a 100 µF electrolytic capacitor and 0.1 µF ceramic capacitor to stabilize. The form factor offers a small and effective user interface while providing a reliable communication link to control the self-righting relief pod.

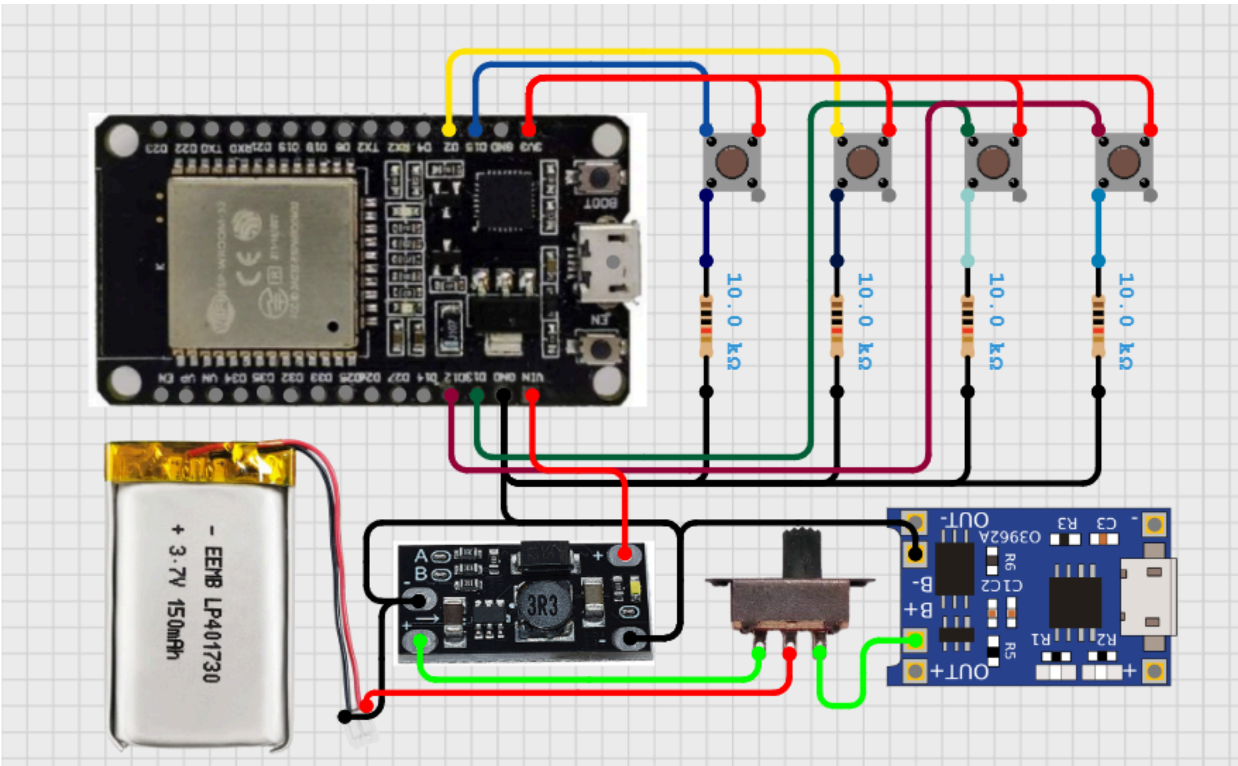

Fig 6: Boat's remote controller circuit diagram

Results and Conclusion

The results of testing align closely with theoretical calculations. The pod's hull submerged to a depth of about 50 mm without added payload. With the 300 g of relief supplies, it submerged to 60 mm deep but formed an efficient pathway for the pod to operate and maintain stability, direction, and depth in all three operational directions. After stability testing, drop tests began to evaluate self-righting performance by engineering trials at various heights, with each height drop being done from a maximum height of 5 ft. Each self-righting process worked successfully and maintained an observable operation. Self-righting performance range can be summarized between −90° and +90°. A full 180° inversion test was not performed as a result of concerns when dropped and/or submerged that onboard equipment could be damaged.

In summary, a successful prototype was designed, developed, and operated in simulated scenarios of a flood. The pod is capable of operating within various water surfaces while carrying payload weights up to 300 g, and dropped from a height of 5 ft. The wireless control system maintained stable communication between the control system from a range of 300 ft. Overall, the prototype was designed from a smaller range. Developments were conducted on pilot studies that indicate the effectiveness and reliability of scaling up the development for larger pods that can provide substantial on-scene assistance during flood-related emergencies.

Future Work

Future enhancements will be focused on upscaling the prototype to carry larger payloads and operate in more extreme flood conditions. Building waterproofing and ruggedizing measures into electronic components will allow for safe tests of extreme drop conditions and full 180° inversion tests. Incorporating autonomous navigation using GPS and obstacle detection systems will reduce the user reliance on manual control and therefore improve deployment in remote areas. Further, energy-saving modes of propulsion, or solar-assisted or hybrid paddle - propeller, will allow for increased endurance of the pods during extended deployments in disaster relief scenes. All of these factors will allow for the necessary transition from a functional prototype to a practical large-scale solution for disaster relief logistics.

References:

[1] Yin Y, Tomita MK, Shavelson RJ. Diagnosing and dealing With student misconceptions: FLOATING and SINKING. Science scope. 2008 Apr 1;31(8).

[2] Akyıldız H, Şimşek C. Self-righting boat design. GİDB Dergi. 2016 Aug 1(06):41-54.

[3] Drobyshevski Y. A note on uprighting of a ship floating upside-down. Ocean engineering. 2004 Aug 1;31(11-12):1447-67.

[4] A Trimulyono et al 2022 IOP Conf. Ser.: Earth Environ. Sci. 1081 012006.